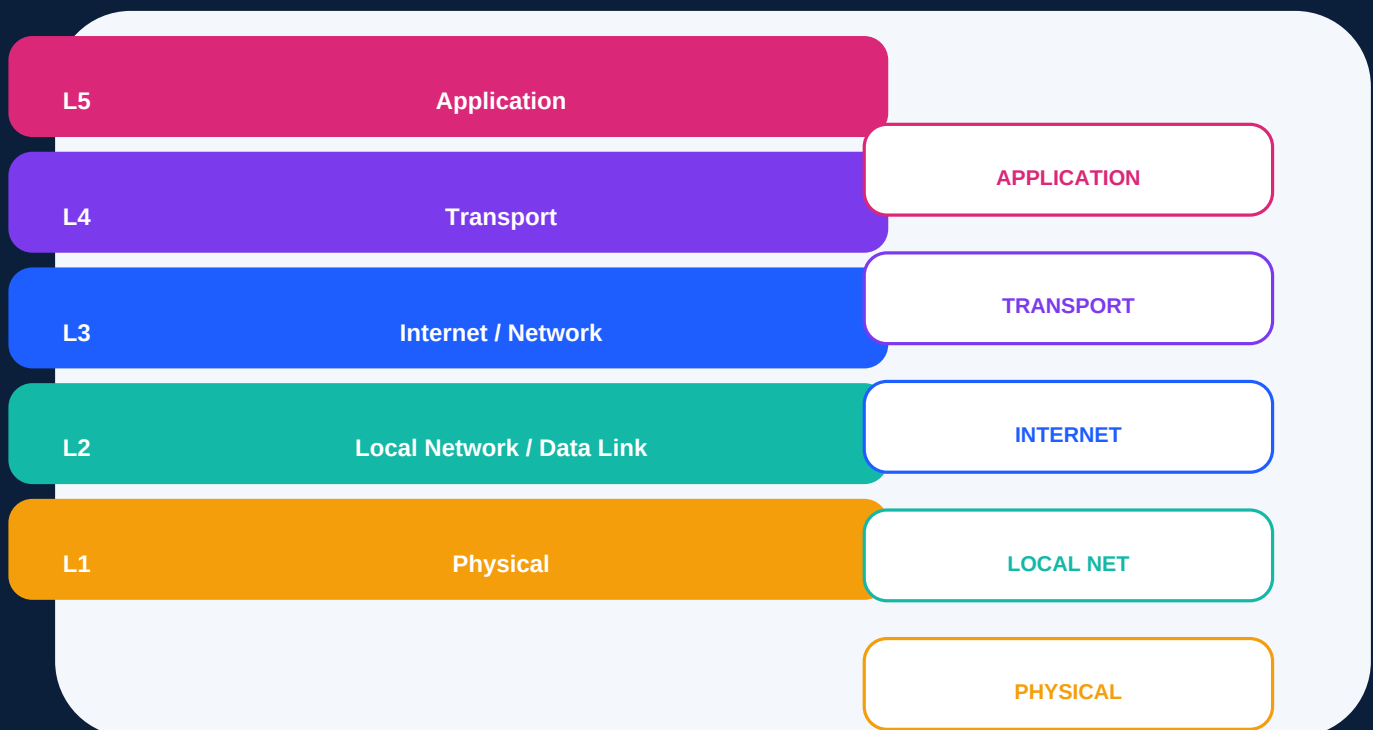


CCNA 200-301

Visual Flashcards

Day 03 - TCP/IP

25 cards | standards | 5-layer model | encapsulation | PDUs



QUESTION

Which organization defines standards used on local area networks?

IEEE

LOCAL AREA NETWORK STANDARDS

Example: Ethernet (802.3)

ANSWER

IEEE (Institute of Electrical and Electronics Engineers)

IEEE defines many LAN technologies, including the Ethernet 802.3 family.

QUESTION

Which organization defines standards used on the Internet?

IETF

INTERNET STANDARDS

Publishes RFC documents

ANSWER

IETF (Internet Engineering Task Force)

IETF develops standards used across the Internet protocol suite.

QUESTION

The IETF publishes standards in documents called [...].

RFC

REQUEST FOR COMMENTS

IETF standards documents

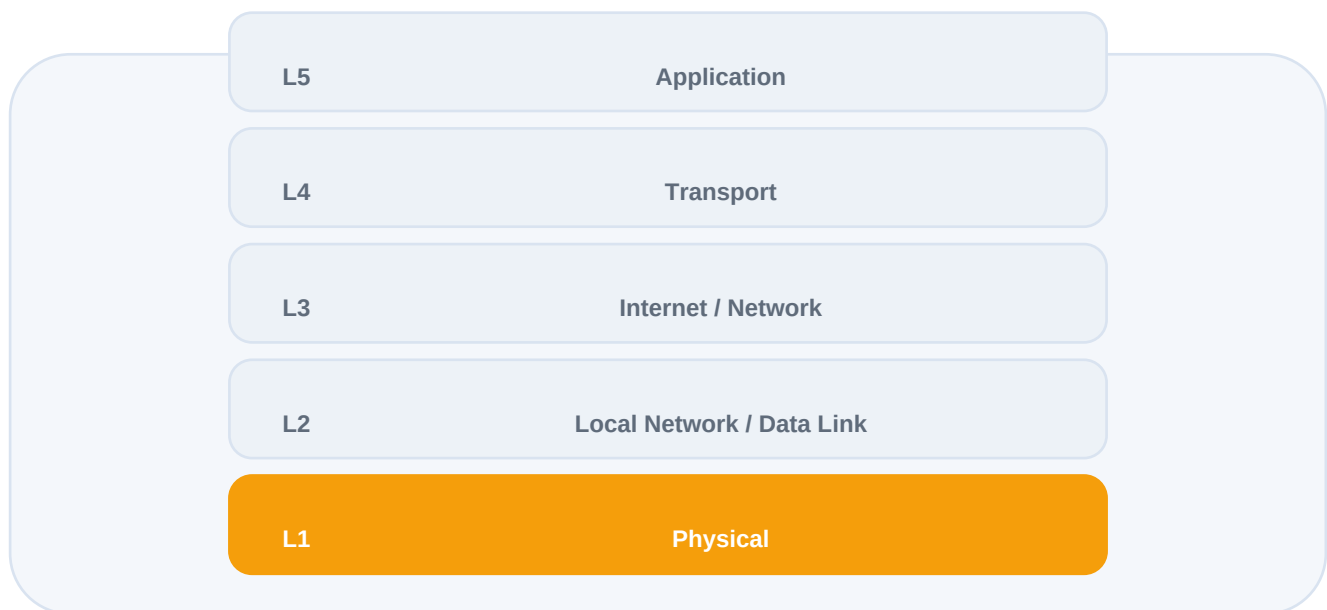
ANSWER

RFCs (Requests for Comments)

RFC means Request for Comments - the document series used by the IETF.

QUESTION

In the 5-layer TCP/IP model, what is Layer 1?



TCP/IP 5-LAYER MODEL

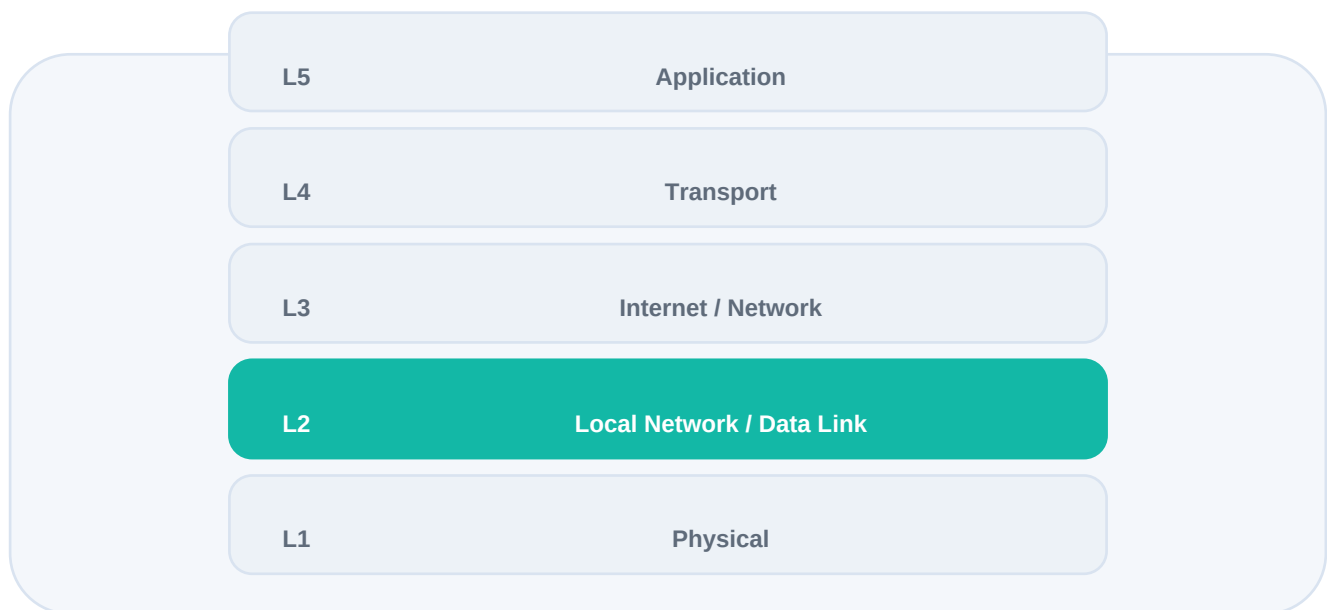
ANSWER

Physical layer

Layer 1 is Physical: media, signaling, and transmission of raw bits.

QUESTION

In the 5-layer TCP/IP model, what is Layer 2?



TCP/IP 5-LAYER MODEL

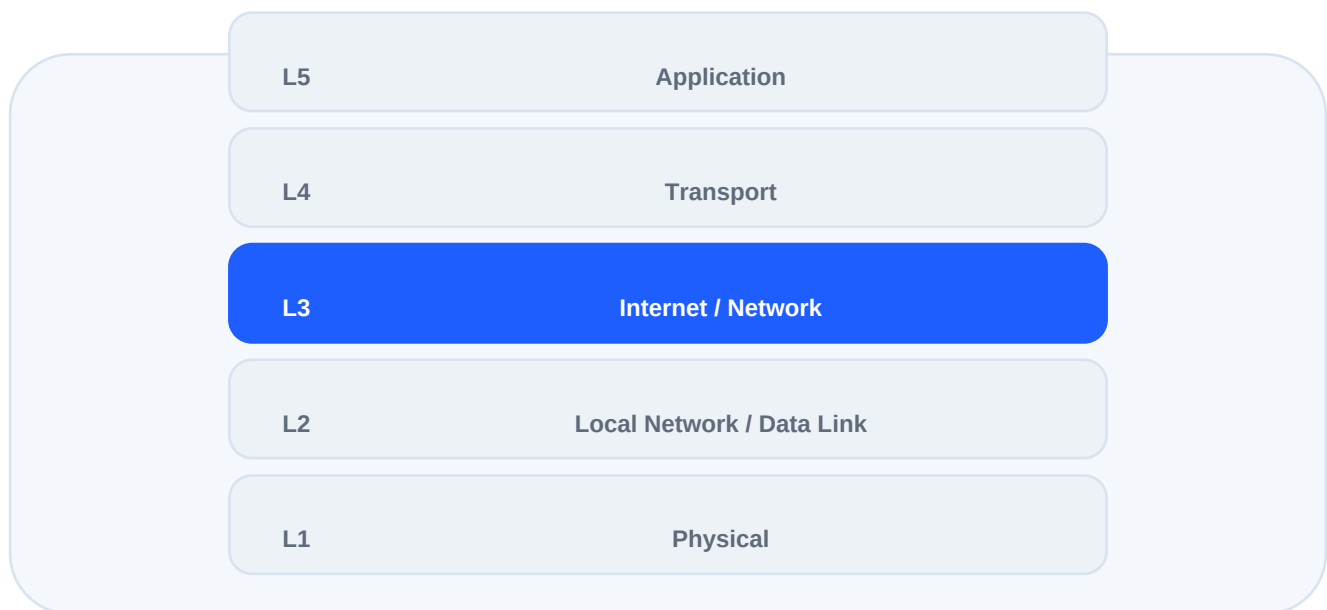
ANSWER

Local Network layer (a.k.a., Data Link layer)

Layer 2 is the Local Network/Data Link layer.

QUESTION

In the 5-layer TCP/IP model, what is Layer 3?



TCP/IP 5-LAYER MODEL

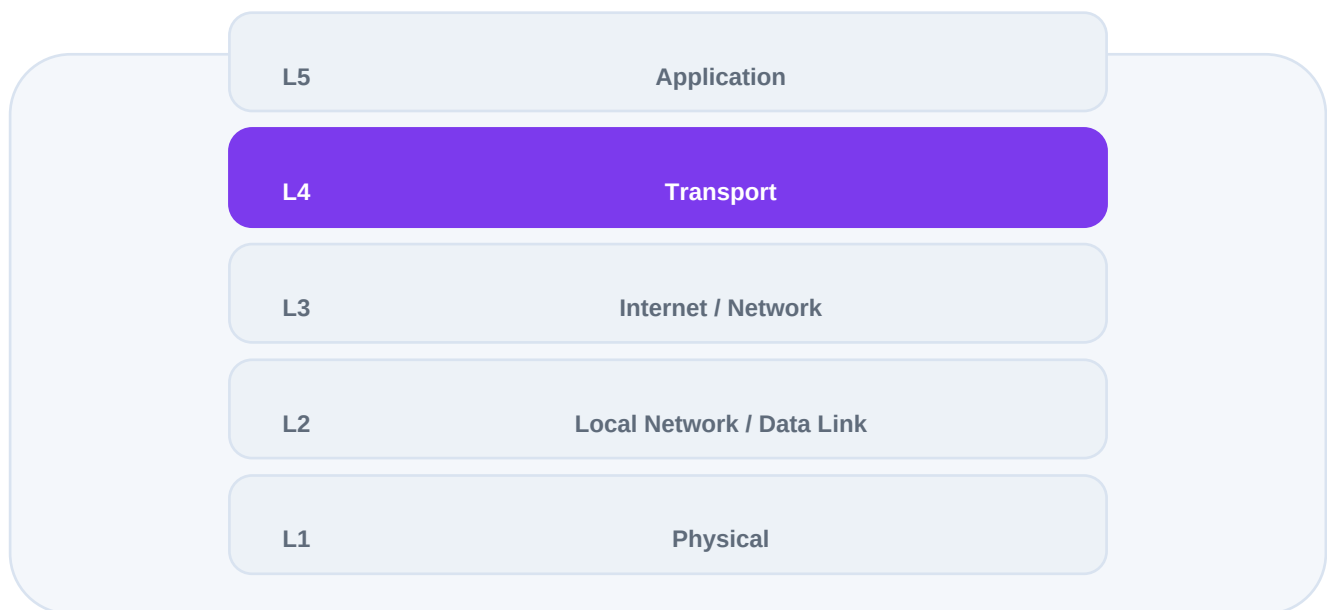
ANSWER

Internet layer (a.k.a., Network layer)

Layer 3 is the Internet/Network layer.

QUESTION

In the 5-layer TCP/IP model, what is Layer 4?



TCP/IP 5-LAYER MODEL

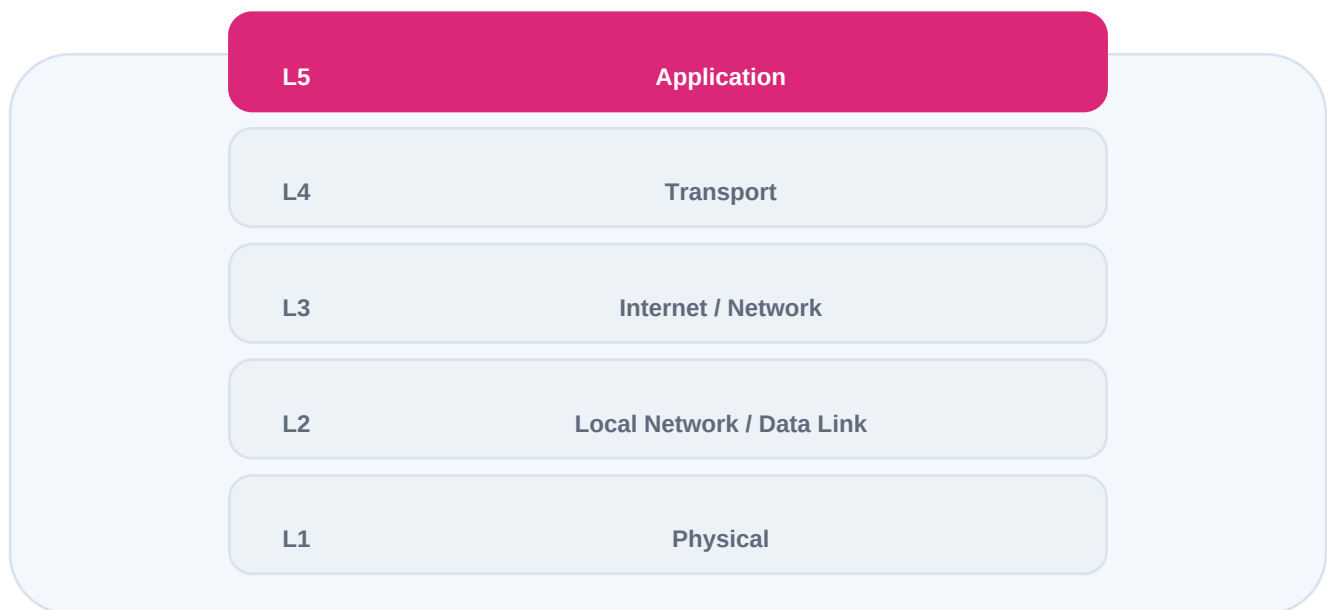
ANSWER

Transport layer

Layer 4 is the Transport layer.

QUESTION

In the 5-layer TCP/IP model, what is Layer 5?



TCP/IP 5-LAYER MODEL

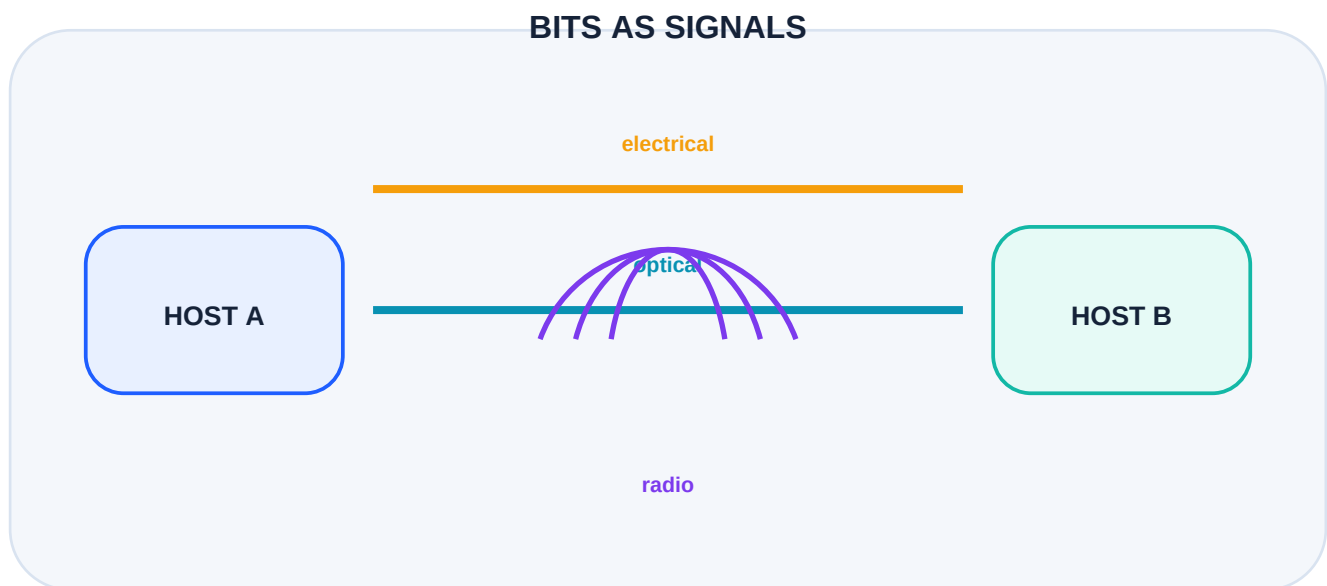
ANSWER

Application layer
***also called Layer 7 due to the OSI model**

Layer 5 is the Application layer; OSI terminology often calls it Layer 7.

QUESTION

In the 5-layer TCP/IP model, which layer sends bits as electrical, optical, or radio signals?



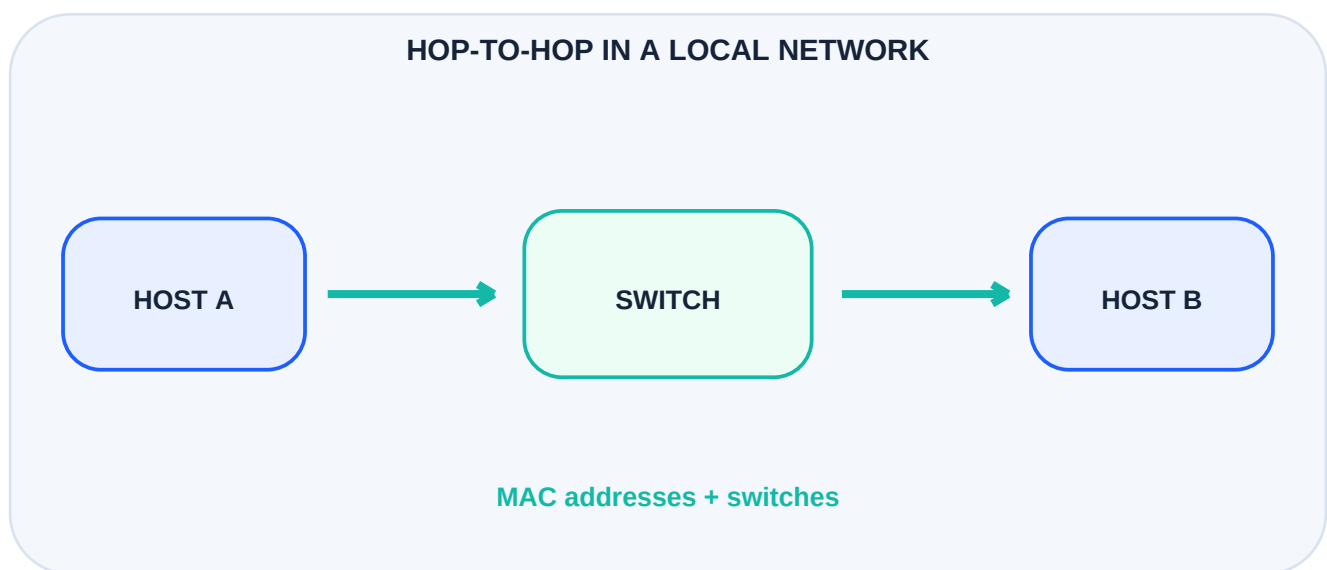
ANSWER

Layer 1 (Physical)

Physical-layer bits travel as electrical, optical, or radio signals.

QUESTION

In the 5-layer TCP/IP model, which layer provides hop-to-hop delivery within a local network using MAC addresses and switches?



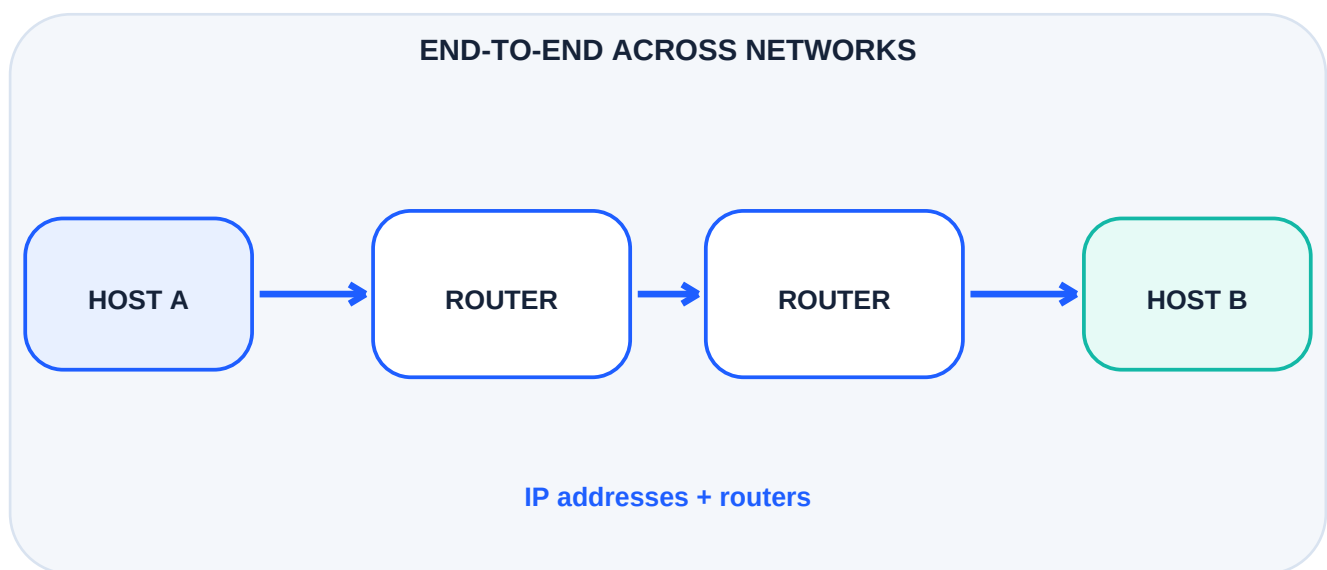
ANSWER

Layer 2 (Local Network layer, a.k.a., Data Link layer)

Layer 2 moves frames hop-by-hop inside a LAN using MAC addresses and switches.

QUESTION

In the 5-layer TCP/IP model, which layer provides end-to-end communication between hosts across networks using IP addresses and routers?



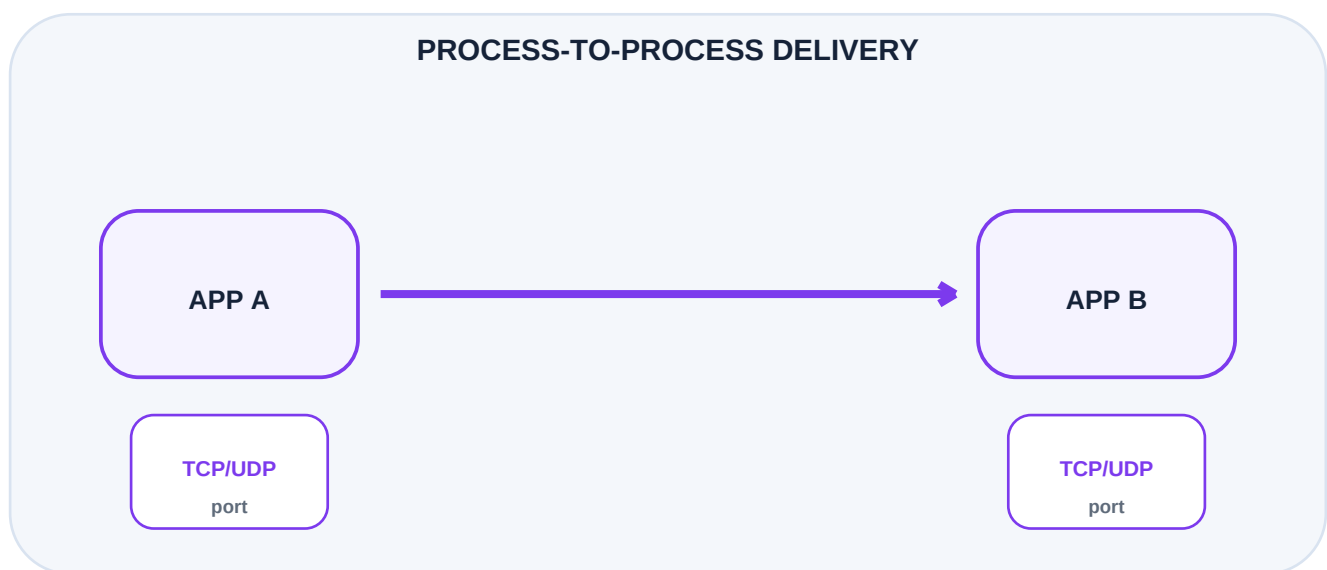
ANSWER

Layer 3 (Internet layer, a.k.a., Network layer)

Layer 3 moves packets between hosts across networks using IP addresses and routers.

QUESTION

In the 5-layer TCP/IP model, which layer provides end-to-end communication between application processes using port numbers?



Port numbers identify application processes

ANSWER

Layer 4 (Transport layer)

Layer 4 identifies application processes with port numbers.

QUESTION

In the 5-layer TCP/IP model, which layer includes protocols for communication between applications, creating and interpreting the data?

APPLICATION PROTOCOLS

HTTP

DNS

SSH

SMTP

Create and interpret application data

ANSWER

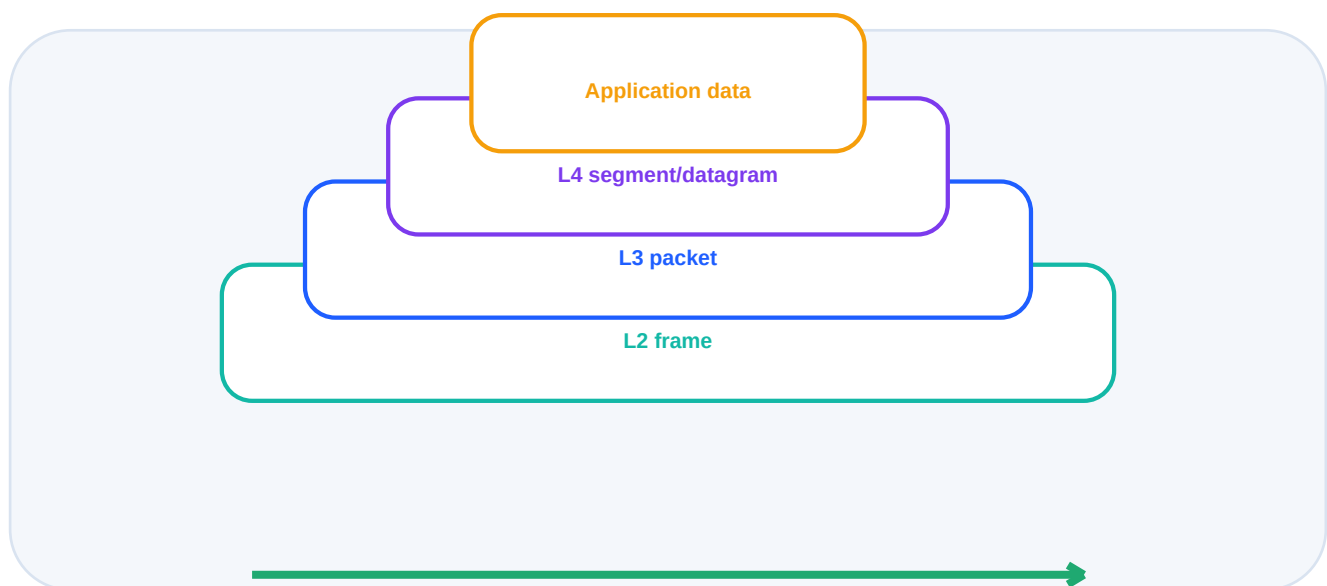
Layer 5 (Application layer)

***Also known as Layer 7 due to the OSI model**

Application-layer protocols create and interpret application data.

QUESTION

What is the name of the process in which headers and a trailer are added to a message before transmitting it?



ADD HEADERS / TRAILER

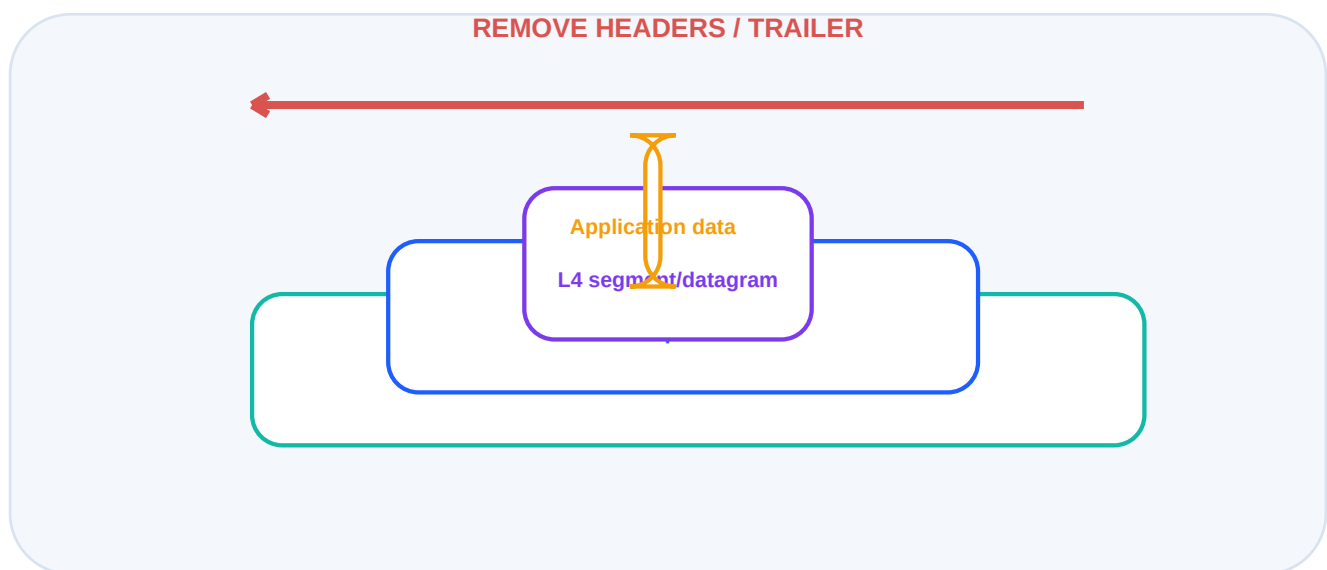
ANSWER

Encapsulation

Encapsulation adds layer information as data moves down the stack before transmission.

QUESTION

What is the name of the process in which headers and a trailer are removed from a message by the receiving host?



ANSWER

Decapsulation (or de-encapsulation)

Decapsulation removes layer information as received data moves up the stack.

QUESTION

What does PDU stand for?

PROTOCOL DATA UNIT

generic name for data at a layer

ANSWER

protocol data unit

PDU is the generic term for a protocol data unit at a particular layer.

QUESTION

An L4PDU is called a [...] or [...]

SEGMENT / DATAGRAM

TCP segment | UDP datagram

ANSWER

**segment or datagram
(segment when using TCP, datagram when
using UDP)**

TCP uses segments, UDP uses datagrams at Layer 4.

QUESTION

An L3PDU is called a [...].

PACKET

Internet / Network layer

ANSWER

packet

The Layer 3 PDU is a packet.

QUESTION

An L2PDU is called a [...].

FRAME

Data Link / Local Network layer

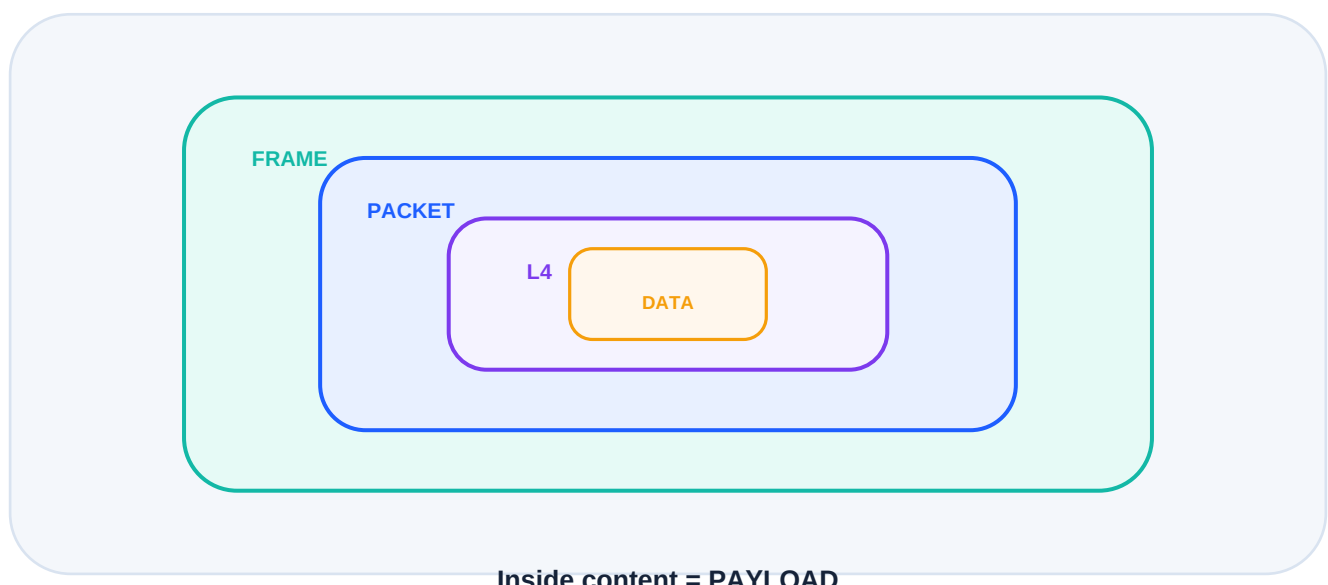
ANSWER

frame

The Layer 2 PDU is a frame.

QUESTION

The contents of each PDU (everything encapsulated by that layer's header/trailer) are called the [...].



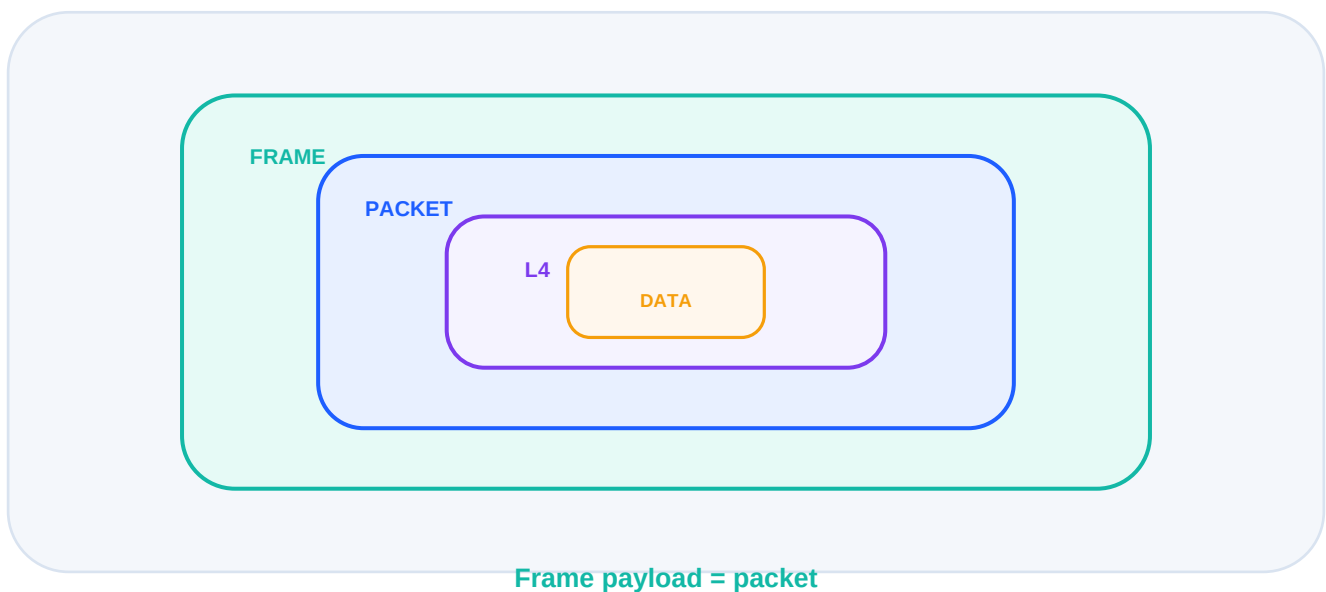
ANSWER

payload

Payload means the content carried inside a layer's header/trailer.

QUESTION

The payload of a frame (L2PDU) is [...].



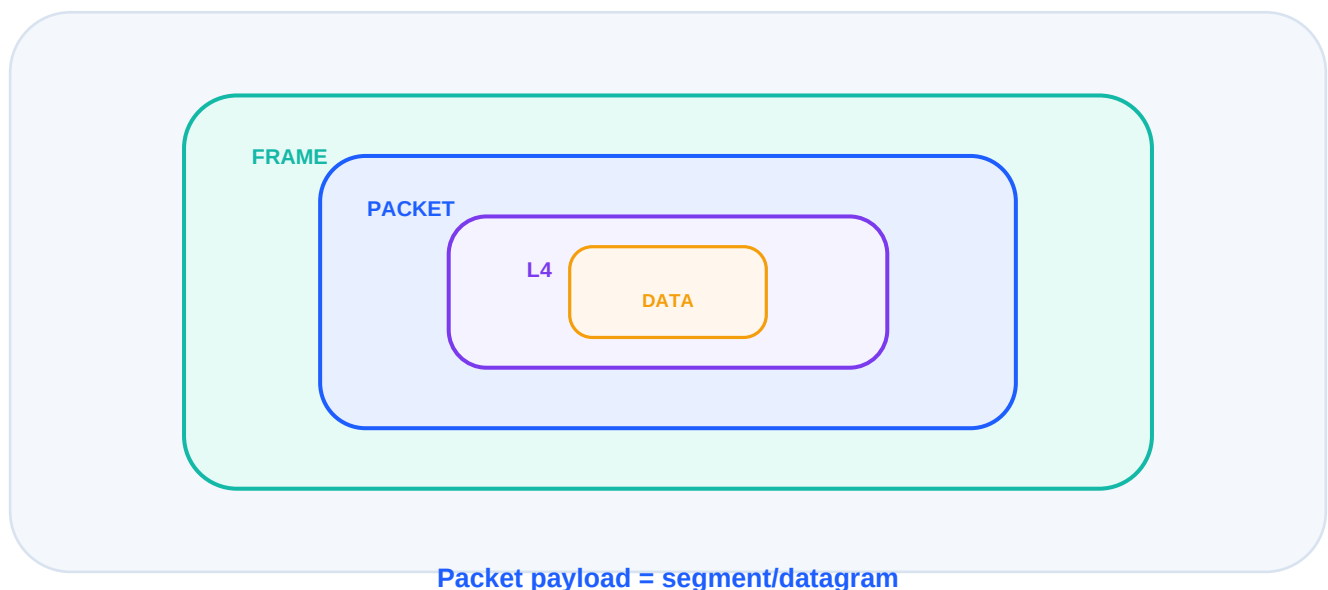
ANSWER

a packet (L3PDU)

A Layer 2 frame carries a Layer 3 packet as its payload.

QUESTION

The payload of a packet (L3PDU) is [...]



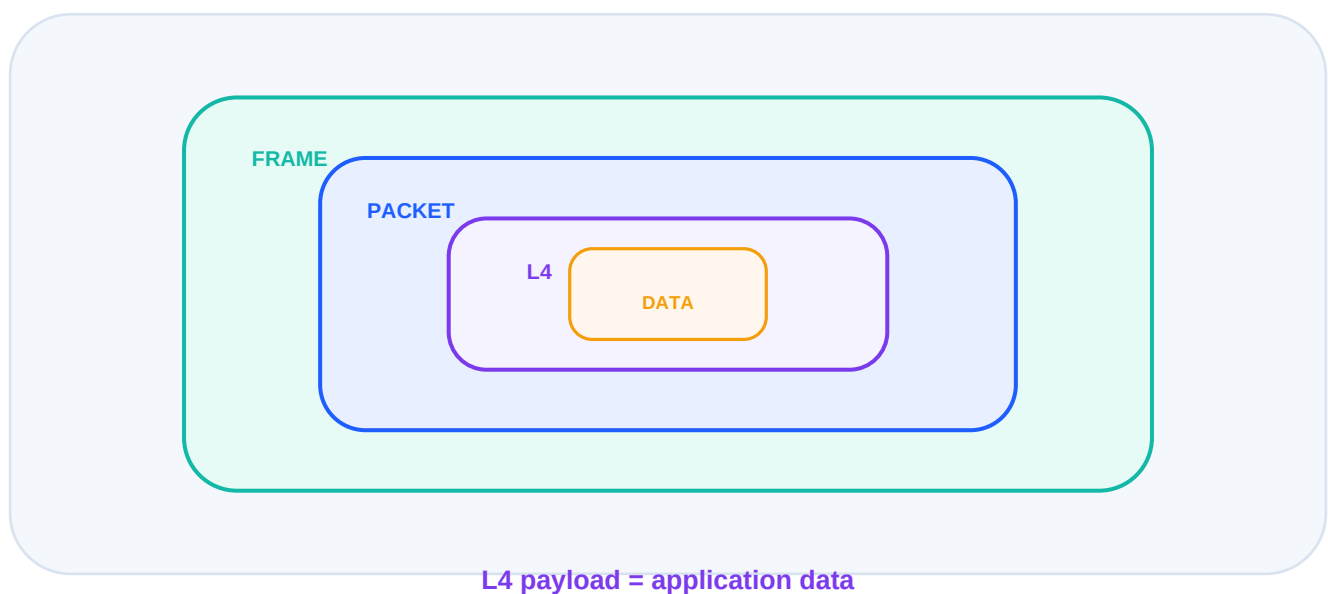
ANSWER

a segment (TCP) or datagram (UDP) (L4PDU)

A Layer 3 packet carries a TCP segment or UDP datagram as its payload.

QUESTION

The payload of a segment or datagram (L4PDU) is [...]



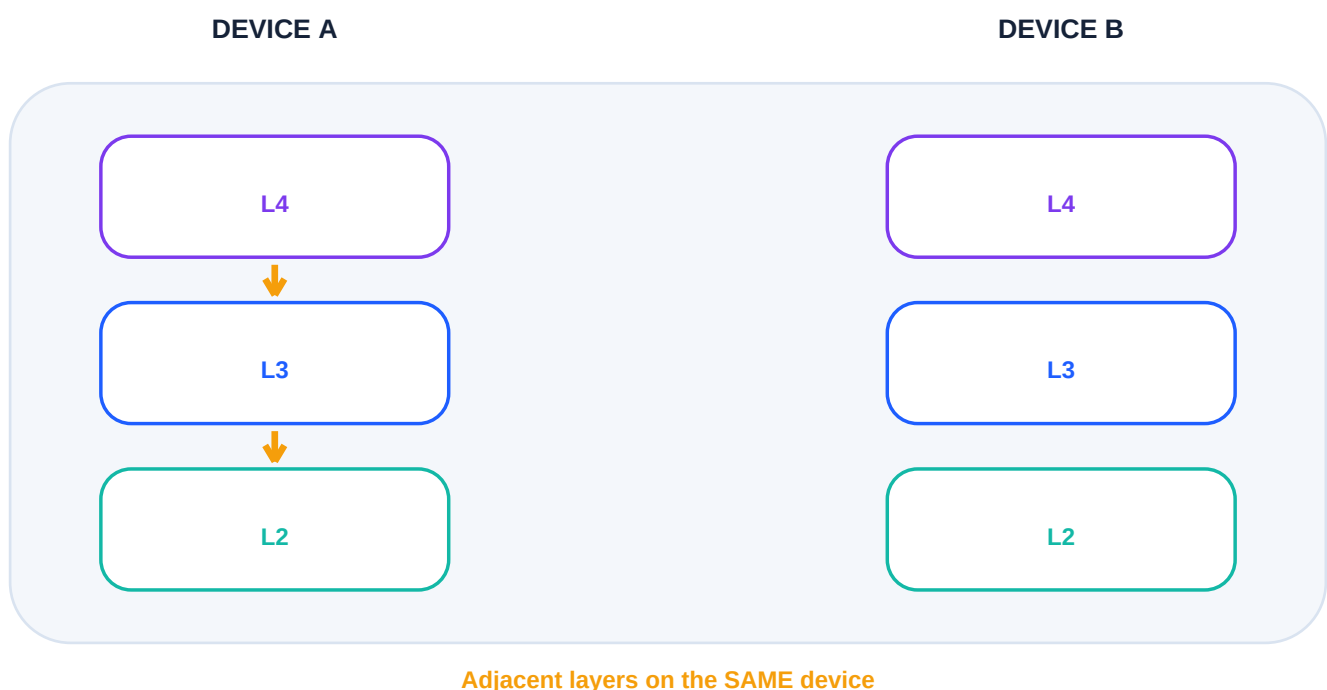
ANSWER

data (from the Application layer)

A Layer 4 segment/datagram carries application data as its payload.

QUESTION

Interaction between a layer and those above and below it is called [...].



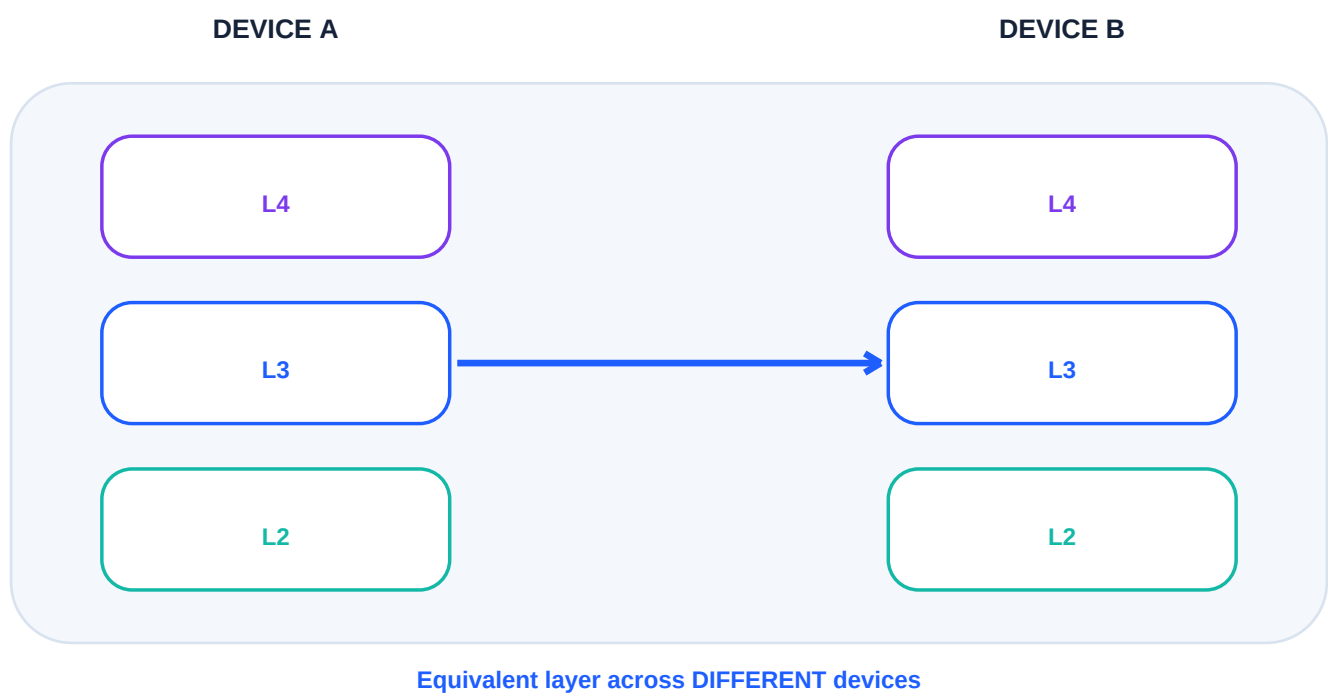
ANSWER

adjacent-layer interaction

Adjacent-layer interaction is vertical interaction between neighboring layers on one device.

QUESTION

Interaction between a layer and the equivalent layer on another device is called [...].



ANSWER

same-layer interaction

Same-layer interaction is logical communication between equivalent layers on different devices.

DAY 03 - QUICK REVIEW

STANDARDS

● IEEE	LAN standards; Ethernet 802.3
● IETF	Internet standards
● RFC	Request for Comments

5-LAYER TCP/IP MODEL

● L5	Application
● L4	Transport
● L3	Internet / Network
● L2	Local Network / Data Link
● L1	Physical

PDU NAMES

● L4	TCP segment / UDP datagram
● L3	Packet
● L2	Frame
● Payload	Content carried inside the PDU

CORE PROCESSES

● Encapsulation	Add headers/trailer before sending
● Decapsulation	Remove headers/trailer when receiving
● Adjacent-layer	Neighboring layers on one device
● Same-layer	Equivalent layers across devices

Encapsulation memory chain

Application data -> L4 segment/datagram -> L3 packet -> L2 frame -> bits on the medium